

The Kv2.2 channel mediates the inhibition of Prostaglandin E2 on glucose-stimulated insulin secretion in pancreatic β -cells

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eLife Assessment

The study presents **valuable** findings on the molecular mechanisms of glucose-stimulated insulin secretion from pancreatic islets, focusing on the main regulatory elements of the signaling pathway in physiological conditions. While the evidence supporting the conclusions is **solid**, the study can be strengthened by the use of a beta cell line or knockout mice. The work will be of interest to cell biologists and biochemists working on diabetes.

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Abstract

Prostaglandin E2 (PGE2) is an endogenous inhibitor of glucose-stimulated insulin secretion (GSIS) and plays an important role in pancreatic β -cell dysfunction in type 2 diabetes mellitus (T2DM). This study aimed to explore the underlying mechanism by which PGE2 inhibits GSIS. Our results showed that PGE2 inhibited Kv2.2 channels via increasing PKA activity in HEK293T cells overexpressed with Kv2.2 channels. Point mutation analysis demonstrated that S448 residue was responsible for the PKA-dependent modulation of Kv2.2. Furthermore, the inhibitory effect of PGE2 on Kv2.2 was blocked by EP2/4 receptor antagonists, while mimicked by EP2/4 receptor agonists. The immune fluorescence results showed that EP1-EP4 receptors are expressed in both mouse and human β -cells. In INS-1(832/13) β -cells, PGE2 inhibited voltage-gated potassium currents and electrical activity through EP2/4 receptors and Kv2.2 channels. Knockdown of Kv2.2 reduced the action potential firing frequency and alleviated the inhibition of PGE2 on GSIS in INS-1(832/13) β -cells. PGE2 impaired glucose tolerance in wild-type mice but did not alter glucose tolerance in Kv2.2 knockout mice. Knockout of Kv2.2 reduced electrical activity, GSIS and abrogated the inhibition of PGE2 on GSIS in mouse islets. In conclusion, we have demonstrated that PGE2 inhibits GSIS in pancreatic β -cells through the EP2/4-Kv2.2 signaling pathway. The findings highlight the

significant role of Kv2.2 channels in the regulation of β -cell repetitive firing and insulin secretion, and contribute to the understanding of the molecular basis of β -cell dysfunction in diabetes.

Introduction

Prostaglandin E2 (PGE2) is the major prostaglandin formed in pancreatic islets and is closely related to islet β -cell dysfunction (Meng et al., 2006 [\[1\]](#); Oshima, Taketo, & Oshima, 2006 [\[2\]](#); Vennemann et al., 2012 [\[3\]](#)). As an endogenous inhibitor of glucose-stimulated insulin secretion (GSIS), PGE2 plays an important role in type 2 diabetes mellitus (T2DM) (Carboneau, Breyer, & Gannon, 2017 [\[4\]](#)). PGE2 has been demonstrated to inhibit GSIS in both mouse models of T2DM and in pancreatic islets obtained from human organ donors with T2DM (Kimple et al., 2013 [\[5\]](#); Neuman et al., 2017 [\[6\]](#); Parazzoli et al., 2012 [\[7\]](#)). Recent studies have shown that plasma levels of PGE2 are correlated with T2DM status, and it has the potential to be a marker of T2DM status (Fenske et al., 2022 [\[8\]](#); Truchan et al., 2021 [\[9\]](#)). However, the underlying mechanism for the PGE2 inhibition of GSIS is not fully understood.

PGE2 functions through the activation of four specific G protein coupled receptor subtypes, termed EP1-4. The EP1 receptor couples to G_q and causes an intracellular Ca^{2+} increase. The EP2 and EP4 receptors couple to G_s to increase intracellular cAMP formation, while the EP3 receptor couples to G_i to decrease intracellular cAMP production (Sugimoto & Narumiya, 2007 [\[10\]](#)). The mRNA expression of all four EP receptors is found in rat, mouse, and human pancreatic islets (Bramswig et al., 2013 [\[11\]](#); Tran, Gleason, & Robertson, 2002 [\[12\]](#); Vennemann et al., 2012 [\[3\]](#)). Among all EP receptors, the EP3 receptor has attracted widespread attention due to its high expression on β -cells in diabetic mouse models (elevated more than 40 times compared to non-diabetic mice) (Kimple et al., 2013 [\[5\]](#)). Currently, it is considered the primary effector for PGE2 in GSIS. Islets from diabetic BTBR mice exhibit increased GSIS when treated with the EP3 antagonist L-798106, and show decreased GSIS after treated with the EP3 agonist PGE1 (Shridas, Zahoor, Forrest, Layne, & Webb, 2014 [\[13\]](#)). However, increasing evidence suggests that EP3 plays a role only in insulin secretion under T2DM (Carboneau et al., 2017 [\[4\]](#)). In human islets from non-diabetic donors, the EP3 antagonist L-798106 does not affect GSIS, but improves insulin secretion in islets from donors with T2DM (Kimple et al., 2013 [\[5\]](#)). Ceddia et al. reported that the EP3 antagonist DG-041 does not affect GSIS in islets from non-diabetic human or wild-type mice on a chow diet, and that global gene knockout of EP3 in mice does not alter GSIS (Ceddia et al., 2016 [\[14\]](#)). The impact of the PGE2 signaling pathway on nondiabetic β -cell function remains unclear. Moreover, the roles of EP1, EP2, and EP4 receptors in GSIS are rarely reported.

Pancreatic β -cells are electrical excitable. GSIS is associated with a complex electrical activity, which is regulated by various voltage-gated plasmalemmal ion channels (Braun et al., 2008 [\[15\]](#); S. N. Yang et al., 2014 [\[16\]](#)). The Kv2 channel family, which consists of Kv2.1 and Kv2.2, plays a crucial role in modulating neural excitability (Liu & Bean, 2014 [\[17\]](#)). The effect of Kv2.1 channels on insulin secretion has been well studied. In rodents, Kv2.1 is recognized as the primary facilitator of delayed rectifier potassium currents in β -cells. Suppressing Kv2.1 results in heightened glucose-stimulated membrane potential amplitude, consequently enhancing GSIS in mouse pancreatic β -cells (Jacobson et al., 2007 [\[18\]](#); MacDonald et al., 2002 [\[19\]](#)). However, Kv2.1 inhibitor stromatoxin shows little effect on human β -cell electrical function or insulin secretion (Braun et al., 2008 [\[15\]](#)), despite the presence of Kv2.1 protein in human islets (Tamarina, Kuznetsov, Fridlyand, & Philipson, 2005 [\[20\]](#)). Increasing evidence suggests that Kv2.1 regulates insulin secretion through channel clusters independently of its electrical function (Dai et al., 2012 [\[21\]](#); Fu et al., 2017 [\[22\]](#); Greitzer-Antes et al., 2018 [\[23\]](#)). Recent studies have shown that human islet β -cells express both Kv2.1 and Kv2.2 channels, with much higher levels of Kv2.2 mRNA expression (Blodgett et al., 2015 [\[24\]](#); Fu et al.,

2017 [\[4\]](#)). Jensen et al. reported that Kv2.2 expression is also involved in GSIS (Jensen et al., 2013 [\[4\]](#)). Nevertheless, the functions of the Kv2.2 channel in the *in vivo* physiology of islet β -cells have not been definitively established.

In this study, we investigated the role of EP1-4 receptors and Kv2.2 in the inhibitory effect of PGE2 on insulin secretion in pancreatic β -cells and Kv2.2 knockout mice. Our findings indicate that: 1) PGE2 reduces Kv2.2 currents via the EP2/4-PKA signaling pathway; 2) PGE2 inhibits β -cell electrical activity through Kv2.2 channels; 3) Knockout of Kv2.2 channels abrogates the PGE2 induced inhibition of GSIS. These results suggest that EP2/4 receptors and Kv2.2 channels play crucial regulatory roles in the normal physiological secretion of insulin.

Results

PGE2 inhibits Kv2.2 currents via the EP2/EP4 signaling pathway in HEK293T cells

We initially conducted experiments to assess the impact of PGE2 on Kv2.2 channels. Kv2.2 channels were overexpressed in HEK293T cells, and Kv2.2 currents were elicited by a 200-ms depolarization pulse from -80 to +40 mV. Extracellular application of 10 μ M PGE2 significantly inhibited Kv2.2 currents. This inhibition was rapid, reaching its maximum effect in approximately 6 minutes (**Fig. 1A and B** [\[4\]](#)). The inhibitory effect of PGE2 on Kv2.2 is dose-dependent, with the maximum effect at around 10 μ M (**Fig. 1C** [\[4\]](#)). Therefore, we used 10 μ M PGE2 for the subsequent experiments. The I-V curve demonstrated that PGE2 significantly inhibited Kv2.2 currents at all positive testing potentials above +20 mV (**Fig. 1D** [\[4\]](#)). Moreover, 10 μ M PGE2 did not alter the steady-state activation and inactivation properties of Kv2.2 currents (**Fig. 1E and F** [\[4\]](#)).

Next, we investigated the mechanism of PGE2 inhibition on Kv2.2 channels. PGE2 functions by activating EP1-4 receptors. We investigated the mRNA and protein expression profiles of EP receptors in HEK293T cells. The mRNAs for all four EP receptors were detected in HEK293T cells, with notably higher levels observed for EP2 and EP4 (**Fig. 2A** [\[4\]](#)). Furthermore, immunofluorescence results confirmed the presence of protein expression for EP1-EP4 receptors in HEK293T cells (**Fig. 2B** [\[4\]](#)). To determine which EP receptor is responsible for the PGE2-induced inhibition, we applied SC51089 (10 μ M, the EP1 receptor antagonist) (Zhou, Qian, Chou, & Iadecola, 2008 [\[4\]](#)), AH6809 (20 μ M, the EP2 receptor antagonist) (Srinivasalu et al., 2020 [\[4\]](#)), L798106 (10 μ M, the EP3 receptor antagonist) (Corboz et al., 2021 [\[4\]](#)), and AH23848 (20 μ M, the EP4 receptor antagonist) (He et al., 2021 [\[4\]](#)) to the extracellular solution to selectively inhibit the respective EP receptors. The acute application of the four antagonists alone had no effect on Kv2.2 currents (**Fig. 2C** [\[4\]](#)). Pretreatment with SC51089 and L798106 for 10 min did not alter the PGE2-induced inhibition. However, pretreatment with AH6809 and AH23848 partially blocked the PGE2-induced inhibition (**Fig. 2D and E** [\[4\]](#)). Furthermore, the EP2 receptor agonist Butaprost (20 μ M) and the EP4 receptor agonist CAY10598 (20 μ M) both inhibited Kv2.2 currents (**Fig. 2F and G** [\[4\]](#)). These results suggest that PGE2 inhibits Kv2.2 via EP2/EP4 signaling pathway.

PGE2 inhibits Kv2.2 currents via the PKA signaling pathway in HEK293T cells

The EP2 and EP4 receptors couple to Gs, increasing intracellular cAMP formation and activating the PKA signaling pathway (Sugimoto & Narumiya, 2007 [\[4\]](#)). We investigated whether PGE2 regulates Kv2.2 via PKA. Treatment with 10 μ M PGE2 rapidly increased PKA activity in HEK293T cells (**Fig. 3A** [\[4\]](#)). The PKA activator, Db-cAMP, could mimic the inhibitory effect of PGE2 on Kv2.2 channels (**Fig. 3B** [\[4\]](#)). The PKA inhibitor, Rp-cAMP, had no effect on Kv2.2 currents (**Fig. 3C** [\[4\]](#)).

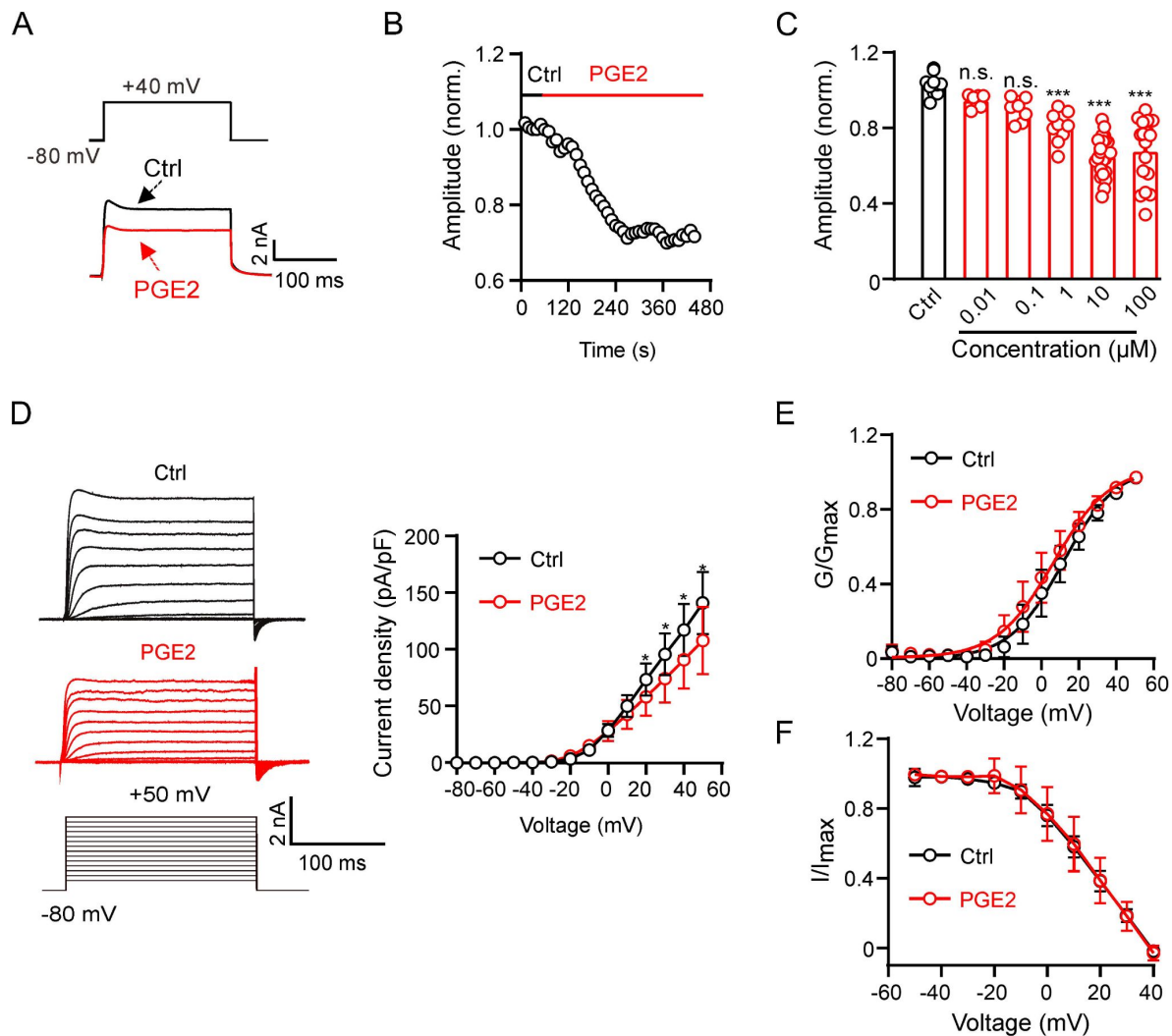


Figure 1.

PGE2 inhibits Kv2.2 channels in HEK293T cells.

(A) Representative Kv2.2 current traces induced by a depolarization pulse from -80 to +40 mV under the control condition (black) and subsequently in the presence of 10 μ M PGE2 (red) in the same HEK293T cell. PGE2 was applied to the extracellular solution and recordings taken at an interval of 10 s. (B) The time course of the Kv2.2 current inhibition by 10 μ M PGE2. (C) PGE2 inhibited Kv2.2 currents in a concentration-dependent manner. n.s., not significant. *** $p < 0.001$. One-way ANOVA with Bonferroni *post hoc* test (0.01 μ M: $n = 7$, $p = 0.4002$; 0.1 μ M: $n = 8$, $p = 0.0672$; 1 μ M: $n = 9$, *** $p = 0.0002$; 10 μ M: $n = 24$, *** $p < 0.0001$; 100 μ M: $n = 17$, *** $p < 0.0001$). (D) Left, representative Kv2.2 current recordings in response to 200-ms 10-mV depolarizing steps from -80 to +50 mV in the control (top, black) and PGE2-treated (bottom, red) groups. Right, plot of the current-voltage relationship from Left ($n = 7$ for each data point). * $p < 0.05$. Two-tailed paired *t*-test. (E) Plot of Kv2.2 current activation curves in the control (black) and PGE2-treated (red, $n = 7$ for each data point) groups. (F) Plot of Kv2.2 current inactivation curves in the control (black) and PGE2-treated (red, $n = 7$ for each data point) groups.

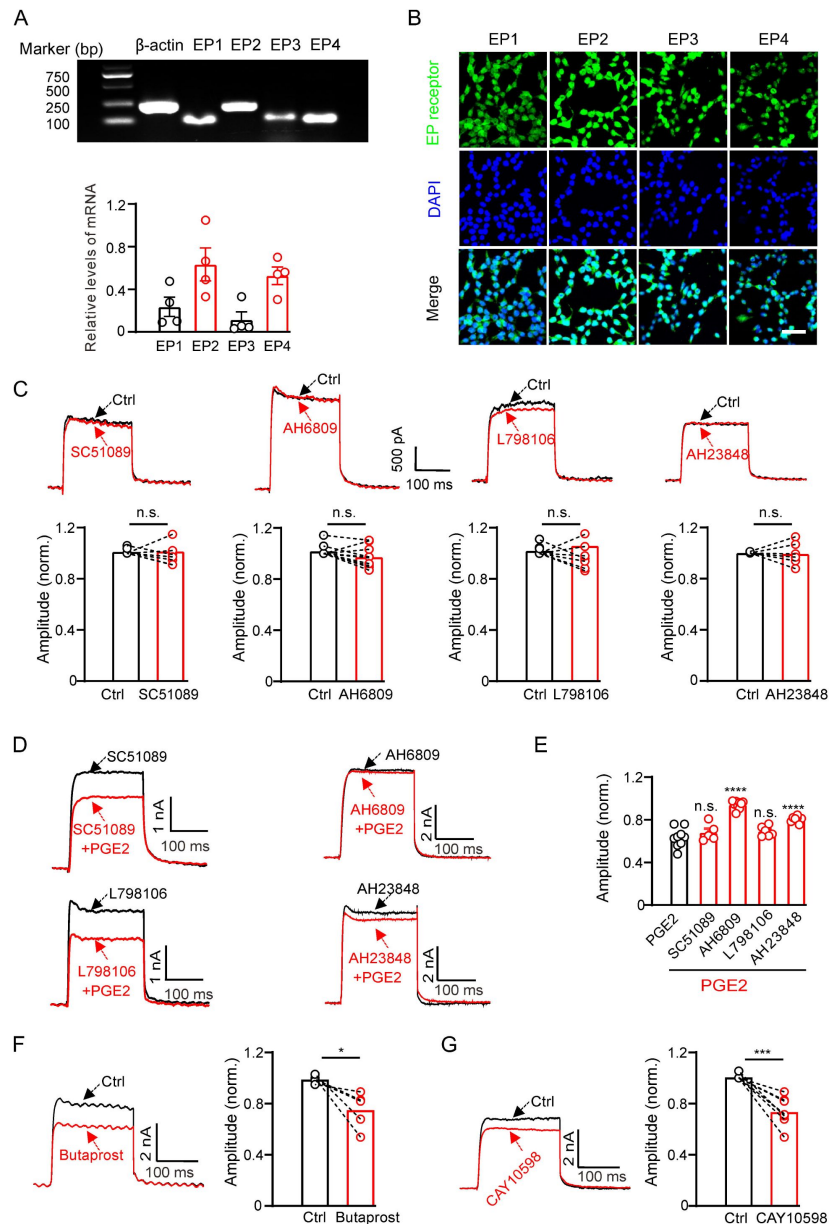


Figure 2.

PGE2 inhibits Kv2.2 currents via the EP2/EP4 signaling pathway in HEK293T cells.

(A) Top, RT-PCR showing the mRNA expression of EP1-4 receptors in HEK293T cells. Bottom, statistics of the mRNA expression of EP1-4 receptors in HEK293T cells ($n = 4$). **(B)** Representative examples of immunofluorescence images showing expression of EP1-4 receptors in HEK293T cells. Scale bar, 20 μ m. **(C)** SC51089 (the EP1 receptor antagonist), AH6809 (the EP2 receptor antagonist), L798106 (the EP3 receptor antagonist), and AH23848 (the EP4 receptor antagonist) per se did not alter Kv2.2 currents. n.s., not significant (SC51089: $n = 8$, $p = 0.9154$; AH6809: $n = 10$, $p = 0.0661$; L798106: $n = 9$, $p = 0.5581$; AH23848: $n = 6$, $p = 0.8827$). **(D)** Representative Kv2.2 current traces induced by a depolarization pulse from -80 to +40 mV in the presence of SC51089, AH6809, L798106 and AH23848 respectively, and subsequently in the presence of an additional 10 μ M PGE2 in the same HEK293T cell. **(E)** Statistical analysis showing the effect of EP1-4 antagonists on PGE2-induced inhibition of Kv2.2 channels. **** $p < 0.001$ versus PGE2 alone by a two-tailed unpaired t -test. n.s., not significant. (+SC51089: $n = 5$, $p = 0.3997$; +AH6809: $n = 10$, **** $p < 0.0001$; +L798106: $n = 6$, $p = 0.1785$; +AH23848: $n = 8$, **** $p < 0.0001$). **(F)** Left, representative Kv2.2 current traces induced by a depolarization pulse from -80 to +40 mV under the control condition and subsequently in the presence of the EP2 receptor agonist Butaprost in the same HEK293T cell. Right, statistics for the amplitude of Kv2.2 currents from Left using a two-tailed paired t -test ($n = 5$, * $p = 0.0306$). **(G)** Similar to F, but with the EP4 receptor agonist CAY10598 in the extracellular solution ($n = 8$, *** $p = 0.0003$).

However, pre-incubation with Rp-cAMP blocked the inhibitory effect of PGE2 on Kv2.2 channels in HEK293T cells (**Fig. 3D**). These findings illustrate that PGE2 inhibits Kv2.2 currents through the PKA signaling pathway.

We further investigated whether PKA regulates Kv2.2 currents via direct phosphorylation of the channel. The Kv2.2 channel protein contains approximately nine putative phosphorylation sites that conform to the minimal consensus sequence for PKA, according to <https://scansite4.mit.edu>. The positions of the nine sites are depicted in **Figure 3E**. To identify the specific site responsible for PKA regulation of Kv2.2 channels, we introduced mutations in the amino acid sequence. We replaced all nine amino acids with aspartic acid to mimic the PKA phosphorylation state of Kv2.2 channels. Among the nine mutations, only the S448D mutation yielded a significant reduction in Kv2.2 currents (**Fig. 3F**). Notably, the S448D mutation also abrogated the inhibitory effect of PGE2 on Kv2.2 (**Fig. 3G**). To further confirm the role of the S448 site in PGE2-induced inhibition of Kv2.2, we also conducted a mutation, changing S448 to alanine (S448A). This mutation effectively prevented the inhibitory effect of PGE2 on Kv2.2 currents (**Fig. 3H**). These data indicate that the S448 site is responsible for the PGE2-induced inhibition of Kv2.2 currents.

PGE2 inhibits endogenous Kv2.2 currents through the EP2/EP4 signaling pathway in INS-1(832/13) β -cells

To investigate the physiological function of PGE2-induced inhibition of Kv2.2, we examined whether PGE2 could also affect the native Kv2.2 currents in pancreatic β -cells. Previous study has demonstrated that Kv2.2 channels contribute to the delayed rectifier outward K^+ currents in pancreatic β -cells (Fu et al., 2017). Consistent with these studies, we observed the protein expression of Kv2.2 channels in INS-1(832/13) β -cells (**Fig. 4A**). The voltage-dependent potassium currents (I_K) in INS-1(832/13) cells were elicited by a 200-ms depolarization pulse from -80 to +40 mV, and the extracellular application of 10 μ M PGE2 significantly inhibited the I_K (**Fig. 4B**).

To assess the contribution of Kv2.2 channels to the PGE2-induced inhibition in INS-1(832/13) cells, we knocked down the expression of Kv2.2 using shRNA. The efficiency of the two shRNAs, named KD1-Kv2.2 and KD2-Kv2.2, were first evaluated in HEK293T cells overexpressing Kv2.2. KD2-Kv2.2 showed higher efficiency than KD1-Kv2.2 and was selected for subsequent experiments (**Fig. 4C**). Knockdown of Kv2.2 by KD2-Kv2.2 significantly reduced endogenous K^+ currents (**Fig. 4D**) in INS-1(832/13) cells, and abrogated PGE2-induced inhibition of K^+ currents (**Fig. 4E and F**). Furthermore, the EP2 receptor agonist Butaprost (20 μ M) and the EP4 receptor agonist CAY10598 (20 μ M) both inhibited the I_K in INS-1(832/13) cells (**Fig. 4G and H**). Pretreatment with TG4155 (1 μ M; the EP2 receptor antagonist) (Jiang et al., 2012) and GW627368 (10 μ M; the EP4 receptor antagonist) (Evans, Howells, Suzuki, Brown, & Cox, 2023) partially blocked the PGE2-induced inhibition, confirming that it involves the same signaling pathway as that in HEK293T cells (**Fig. 4I and J**).

PGE2 reduces electrical activity via Kv2.2 channels in INS-1(832/13) cells

GSIS is associated with β -cell electrical activity, which is regulated by Kv2 potassium channels (Drews, Krippeit-Drews, & Dufer, 2010). Therefore, we tested whether PGE2 affects β -cell electrical activity. We found that 10 μ M PGE2 reduced action potential (AP) frequency and increased AP half-width in INS-1(832/13) cells with little effect on the peak amplitude of AP (**Fig. 5A and B**). Furthermore, knockdown of Kv2.2 with Kv2.2-specific shRNA abrogated PGE2-induced effect on INS-1(832/13) cells (**Fig. 5C**). This suggests that PGE2 reduces electrical activity via Kv2.2 channels in INS-1(832/13) cells. Since PGE2 inhibits Kv2.2 via EP2/4 receptors, it is plausible to hypothesize that PGE2 reduces electrical activity via EP2/4 receptors in INS-1(832/13) cells. To test the hypothesis, we first investigated the expression of EP receptors in INS-1(832/13) cells. Immunofluorescence results revealed that all EP1-EP4 receptors are expressed in INS-1(832/13)

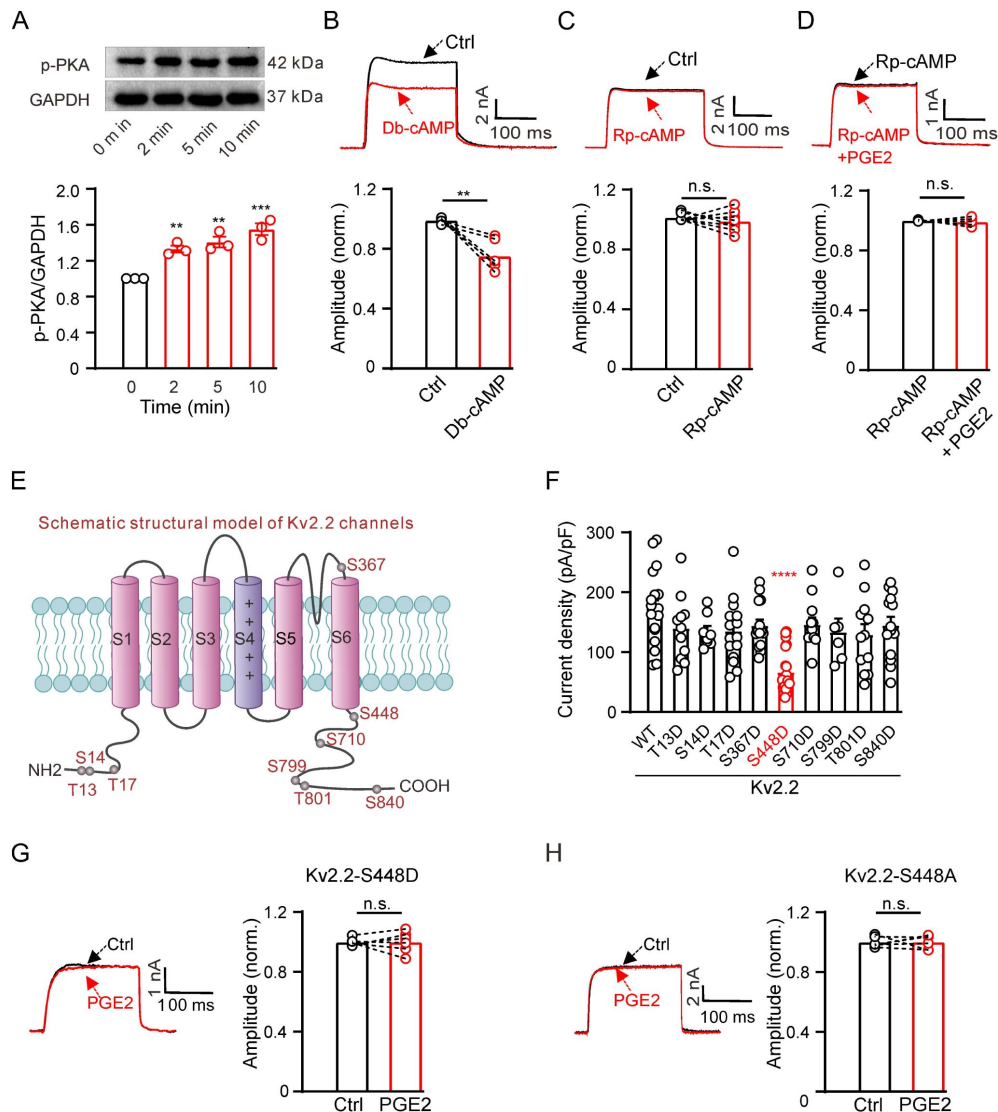


Figure 3.

PGE2 inhibits Kv2.2 currents via the PKA signaling pathway.

(A) Top, representative western blot showing the PKA phosphorylation level in HEK293T cells following treatment with 10 μ M PGE2 for time intervals of 2, 5, and 10 minutes. Bottom, statistics from three independent experiments using a one-way ANOVA with Bonferroni *post hoc* test ($n = 3$; 2 min: $**p = 0.0078$; 5 min: $**p = 0.002$; 10 min: $***p = 0.0002$). **(B)** Top, representative Kv2.2 current traces induced by a depolarization pulse from -80 to +40 mV under the control condition (black) and subsequently in the presence of 10 μ M Db-cAMP (red) in the same HEK293T cell. Bottom, statistics for the amplitude of Kv2.2 currents from Top using a two-tailed paired *t*-test ($n = 6$, $**p = 0.0035$). **(C)** Similar to B, but with Rp-cAMP in the extracellular solution ($n = 8$, $p = 0.3851$). **(D)** Top, representative Kv2.2 current traces induced by a depolarization pulse from -80 to +40 mV in the presence of 10 μ M Rp-cAMP (black) and subsequently in the presence of an additional 10 μ M PGE2 (red) in the same HEK293T cell. Bottom, statistics for the amplitude of Kv2.2 current from Top using a two-tailed paired *t*-test ($n = 5$, $p = 0.6425$). n.s., not significant. **(E)** Schematic structural models of Kv2.2 channels indicate the positions of nine potential PKA phosphorylation sites. **(F)** Statistics for the amplitude of wild-type and various mutant Kv2.2 channel currents, induced by a depolarization pulse from -80 to +40 mV ($n = 6-24$). $****p < 0.0001$ compared to wild-type Kv2.2 by a one-way ANOVA with Bonferroni *post hoc* test. **(G)** Left, representative Kv2.2-S448D mutant channel current traces induced by a depolarization pulse from -80 to +40 mV under the control condition (black) and subsequently in the presence of 10 μ M PGE2 (red) in the same HEK293T cell. Right, statistics for the amplitude of Kv2.2 current from Left using a two-tailed paired *t*-test ($n = 6$, $p > 0.9999$). n.s., not significant. **(H)** Similar to G, this section presents data for the Kv2.2-S448A mutant channel in HEK293T cells ($n = 6$, $p > 0.9999$).

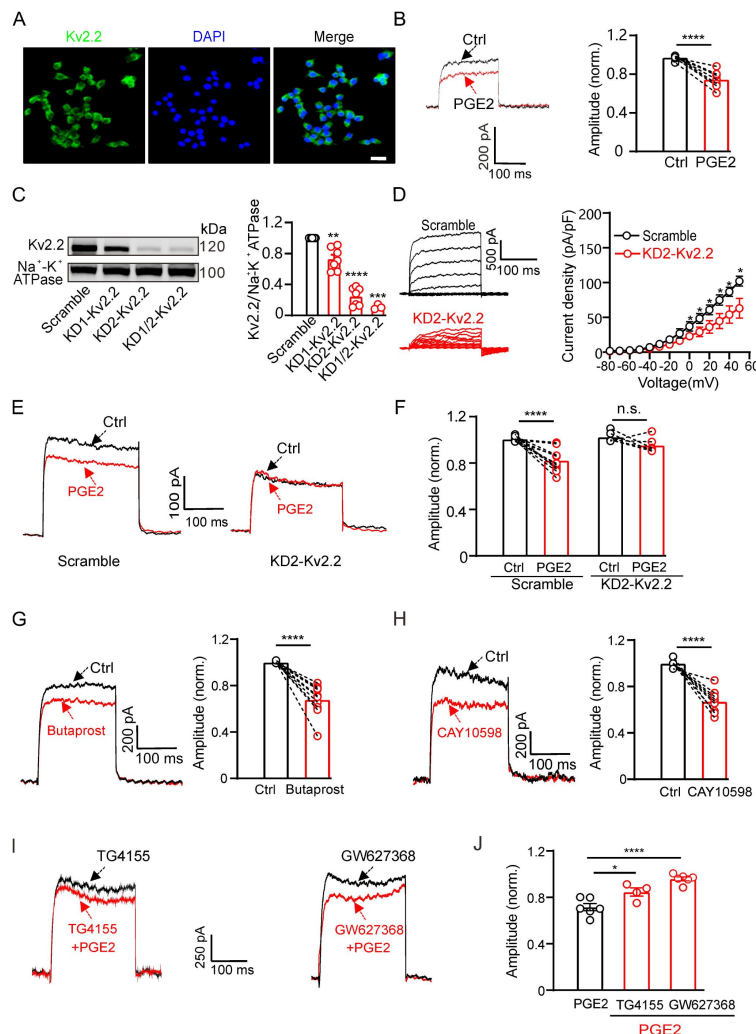


Figure 4.

PGE2 inhibits native Kv2.2 channels in INS-1(832/13) cells.

(A) Representative examples of immunofluorescence images showing high expression of Kv2.2 channels in INS-1(832/13) cells. Scale bar, 20 μ m. (B) Left, representative I_K traces induced by a depolarization pulse from -80 to +40 mV under the control condition (black) and subsequently in the presence of 10 μ M PGE2 (red) in the same INS-1(832/13) cell. Right, statistics for the amplitude of I_K from Left using a two-tailed paired t -test ($n = 10$, **** $p < 0.0001$). (C) Left, representative examples of western blot images showing the effects of the two shRNA oligos (KD1-Kv2.2 and KD2-Kv2.2) on Kv2.2 channel surface expression in HEK293T cells transfected with Kv2.2. Right, statistics for cell surface expression of Kv2.2 channels from Left using a two-tailed unpaired t -test. KD1-Kv2.2: $n = 7$, ** $p = 0.0066$; KD2-Kv2.2: $n = 7$, **** $p < 0.0001$; KD1/KD2-Kv2.2: $n = 3$, *** $p = 0.0005$, compared with scramble. (D) Knockdown of Kv2.2 channels significantly reduced I_K amplitude in INS-1(832/13) cells ($n = 6$, * $p < 0.05$). (E) Left, representative I_K traces induced by a depolarization pulse from -80 to +40 mV under the control condition (black) and subsequently in the presence of 10 μ M PGE2 (red) in the same INS-1(832/13) cell transfected with scramble control or KD2-Kv2.2. (F) Knockdown of Kv2.2 channels abrogated the inhibitory effect of PGE2 on I_K in INS-1(832/13) cells. Scramble-PGE2: $n = 11$, **** $p < 0.0001$; KD2-PGE2: $n = 6$, $p = 0.1227$; two-tailed paired t -test. (G) Left, representative Kv2.2 current traces induced by a depolarization pulse from -80 to +40 mV under the control condition and subsequently in the presence of the EP2 receptor agonist Butaprost in the same INS-1(832/13) cell. Right, statistics for the amplitude of Kv2.2 currents from Left using a two-tailed paired t -test ($n = 10$, **** $p < 0.0001$). (H) Similar to G, but with the EP4 receptor agonist CAY10598 in the extracellular solution ($n = 9$, **** $p < 0.0001$). (I) Representative Kv2.2 current traces induced by a depolarization pulse from -80 to +40 mV in the presence of TG4155 or GW627368, followed by the addition of 10 μ M PGE2 in the same INS-1(832/13) cell. (J) Statistical analysis showing the effects of EP2 and EP4 antagonists on PGE2-induced inhibition of Kv2.2 channels. * $p < 0.05$ and **** $p < 0.0001$ versus PGE2 alone by a two-tailed unpaired t -test. (+TG4155: $n = 4$, * $p = 0.0242$; + GW627368: $n = 6$, **** $p < 0.0001$).

cells (**Fig. 5D**). In addition, both mouse and human islet β -cells were found to express all four EP receptors (**Fig. 5E**). As expected, the EP2 receptor agonist Butaprost (20 μ M) and the EP4 receptor agonist CAY10598 (20 μ M) both inhibited AP frequency in INS-1(832/13) cells (**Fig. 6A and B**). This indicates that PGE2 reduces electrical activity via EP2/4 in INS-1(832/13) cells.

PGE2 regulates insulin secretion through Kv2.2 channels in INS-1(832/13) cells and mice

We next investigated whether Kv2.2 regulates the effect of PGE2 on GSIS in INS-1(832/13) cells. As expected, 10 μ M PGE2 significantly reduced GSIS with little effect on basal insulin secretion in INS-1(832/13) cells (**Fig. 7A**). Knockdown of Kv2.2 with KD2 shRNA did not alter basal insulin secretion but reduced GSIS in INS-1(832/13) cells (**Fig. 7B**). Moreover, knockdown of Kv2.2 with KD2 shRNA greatly alleviated the inhibitory effect of PGE2 on GSIS (percent inhibition: scramble, 62%; KD2, 14%, **Fig. 7B**). To further validate that PGE2 regulates GSIS through PKA-mediated phosphorylation of the KV2.2 channel, we overexpressed the Kv2.2-S448A mutant channel in INS-1(832/13) cells. Compared to cells transfected with either an empty vector (as a control) or the wild-type Kv2.2 plasmid, overexpression of Kv2.2-S448A significantly attenuated the inhibitory effect of PGE2 on GSIS (**Fig. 7C**).

To investigate the physiological role of Kv2.2 during GSIS *in vivo*, a mouse model was utilized in which the Kv2.2 coding gene is disrupted at exon 2 (Supplemental Fig. S1A). The targeting cassette removes 446 bp in exon 2 of the Kv2.2 coding gene sequence. The disrupted sequence is detected using PCR to produce one amplicon with one primer inside the targeting sequence in combination with one Kv2.2 specific primer outside the targeting sequence (Supplemental Fig. S1A). The Kv2.2 protein was detected only in wild-type islets using a Kv2.2-specific antibody (Supplemental Fig. S1B). The body weight of the Kv2.2 knockout mice was similar to that of the wild-type littermates (Supplemental Fig. S1C). Kv2.2^{-/-} mice were assessed for possible impairment in glucose homeostasis using an intraperitoneal glucose tolerance test. Kv2.2^{-/-} mice had similar fasting blood glucose levels with control group (**Fig. 7D**). PGE2 treatment worsened glucose tolerance in control animals but had little effect on Kv2.2^{-/-} mice (**Fig. 7D**). This suggests that PGE2 modulates GSIS *in vivo* through Kv2.2 channels. The Kv2.2 gene knockout mice used in this study are global knockouts. To further confirm the role of Kv2.2 in insulin secretion, we isolated pancreatic islets for GSIS experiments. Islets from Kv2.2^{-/-} mice exhibited reduced GSIS compared to islets from wild-type mice (**Fig. 7E**). While PGE2 reduced GSIS in islets from wild-type mice, it had little effect on GSIS in islets from Kv2.2^{-/-} mice (**Fig. 7E**).

PGE2 reduces electrical activity via Kv2.2 channels in mouse β -cells within islets

Finally, we investigated the effect of PGE2 on the electrical activity of β -cells within islets. The I_K in Kv2.2^{-/-} β -cells showed significantly reduced amplitudes (**Fig. 8A**). Similar to our observations in INS-1(832/13) cells above, 10 μ M PGE2 inhibited I_K in mouse β -cells, and Kv2.2 ablation abolished the inhibitory effect of PGE2 (**Fig. 8B**). PGE2 reduced the firing frequency of AP induced by 20 mM glucose in β -cells within islets, while had little effect on the AP peak amplitude and half-width (**Fig. 8C**). Furthermore, the effect of PGE2 on mouse islet β -cell electrical activity was abrogated by knockout of Kv2.2 (**Fig. 8D**).

Discussion

While EP3 has been demonstrated to exert an inhibitory influence on GSIS, increasing evidence suggests that the impact of EP3 may manifest primarily when β -cell dysfunction is already established, as observed in T2DM (Carboneau et al., 2017; Truchan et al., 2021). In our current

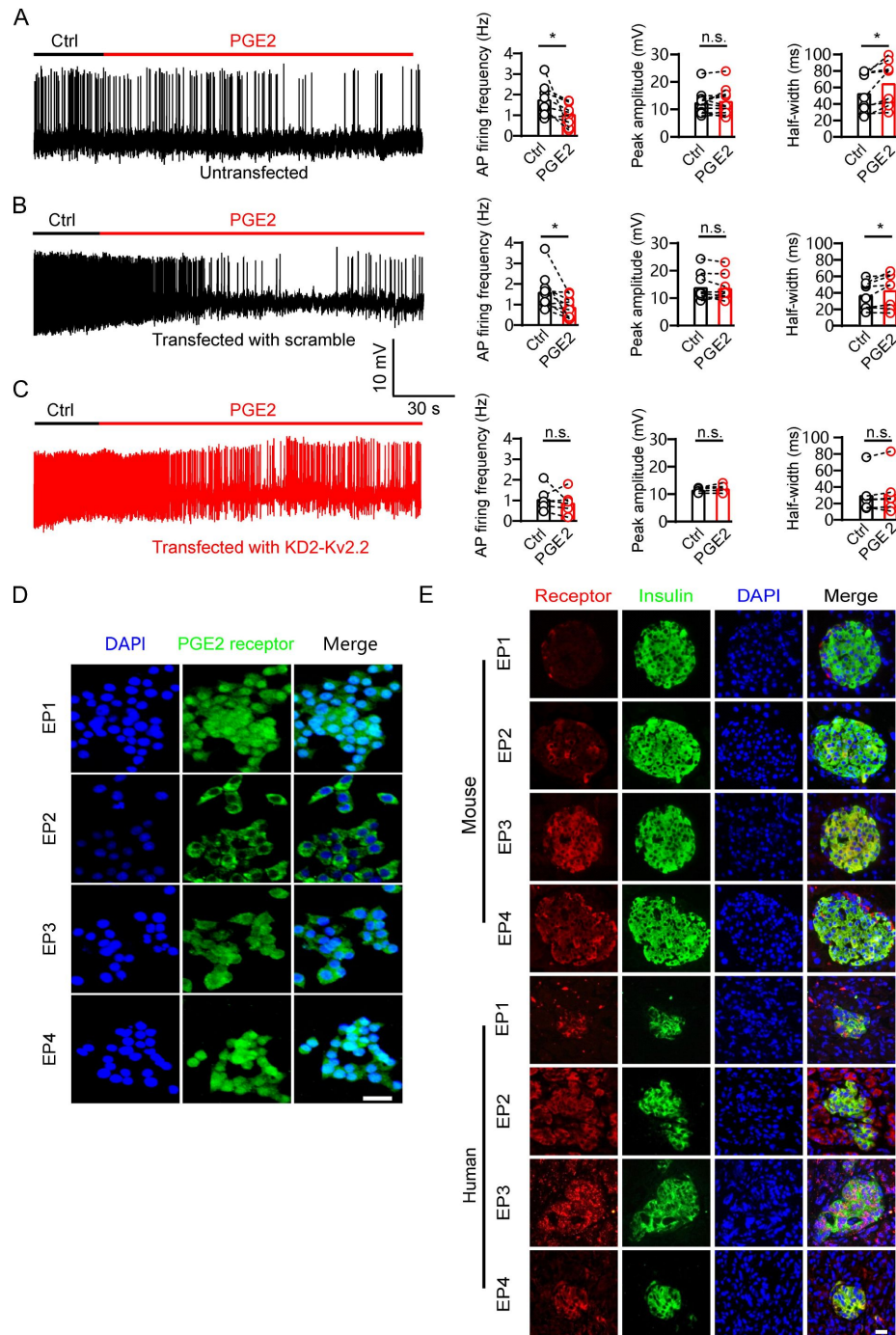


Figure 5.

PGE2 reduces β -cell electrical excitability through Kv2.2 channels.

(A) Left, representative action potential (AP) firings induced by 20 mM glucose under the control condition and subsequently in the presence of 10 μ M PGE2 in the same INS-1(832/13) cell. Right, statistics for the AP firing frequency (* p = 0.0385), amplitude (p = 0.9478), and half-width (* p = 0.0101) from A (n = 9). Two-tailed paired t -test. (B) Similar to A, but the INS-1(832/13) cells were transfected with scramble-Kv2.2 (n = 8) (frequency: * p = 0.0285; amplitude: p = 0.6603; half-width: * p = 0.0281). (C) Similar to B, but the INS-1(832/13) cells were transfected with KD2-Kv2.2 (n = 6). n.s., not significant (frequency: p = 0.4564; amplitude: p = 0.1601; half-width: p = 0.3034). (D) Representative immunofluorescence images showing expression of EP1-4 receptors in INS-1(832/13) cells. Scale bar, 20 μ m. (E) Representative immunofluorescence images showing expression of EP1-4 receptors in mouse and human islets. Scale bar, 20 μ m.

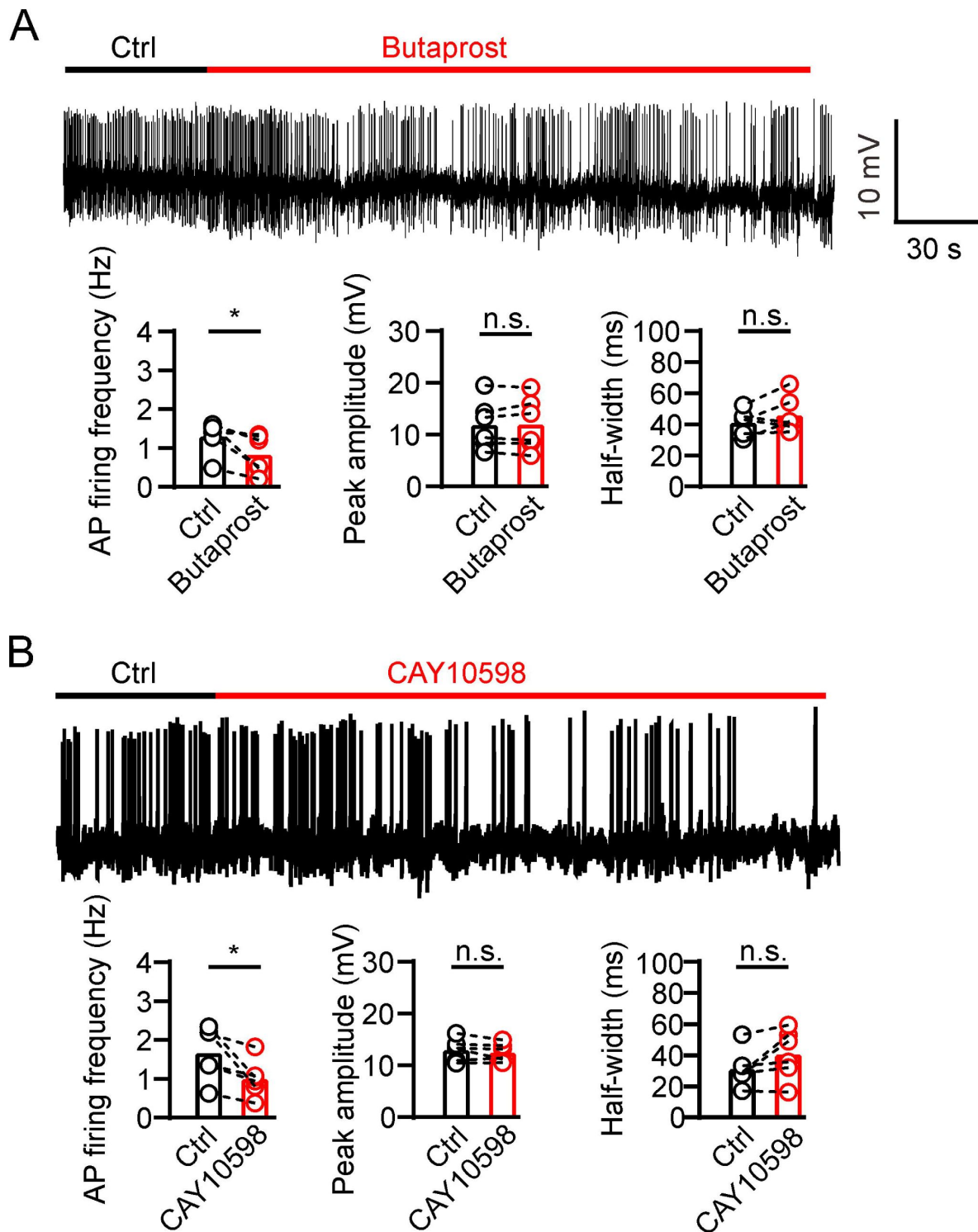


Figure 6.

PGE2 reduces β -cell electrical excitability through EP2/4.

(A) Top, representative AP firings induced by 20 mM glucose under the control condition and subsequently in the presence of 20 μ M Butaprost in the same INS-1(832/13) cell. Bottom, statistics for the AP firing frequency (* p = 0.0312), amplitude (p = 0.7354), and half-width (p = 0.2067) from Top (n = 6). Two-tailed paired t -test. (B) Similar to A, but in the presence of 20 μ M CAY10598 (n = 6), AP firing frequency (* p = 0.0291), amplitude (p = 0.2211), half-width (p = 0.0753).

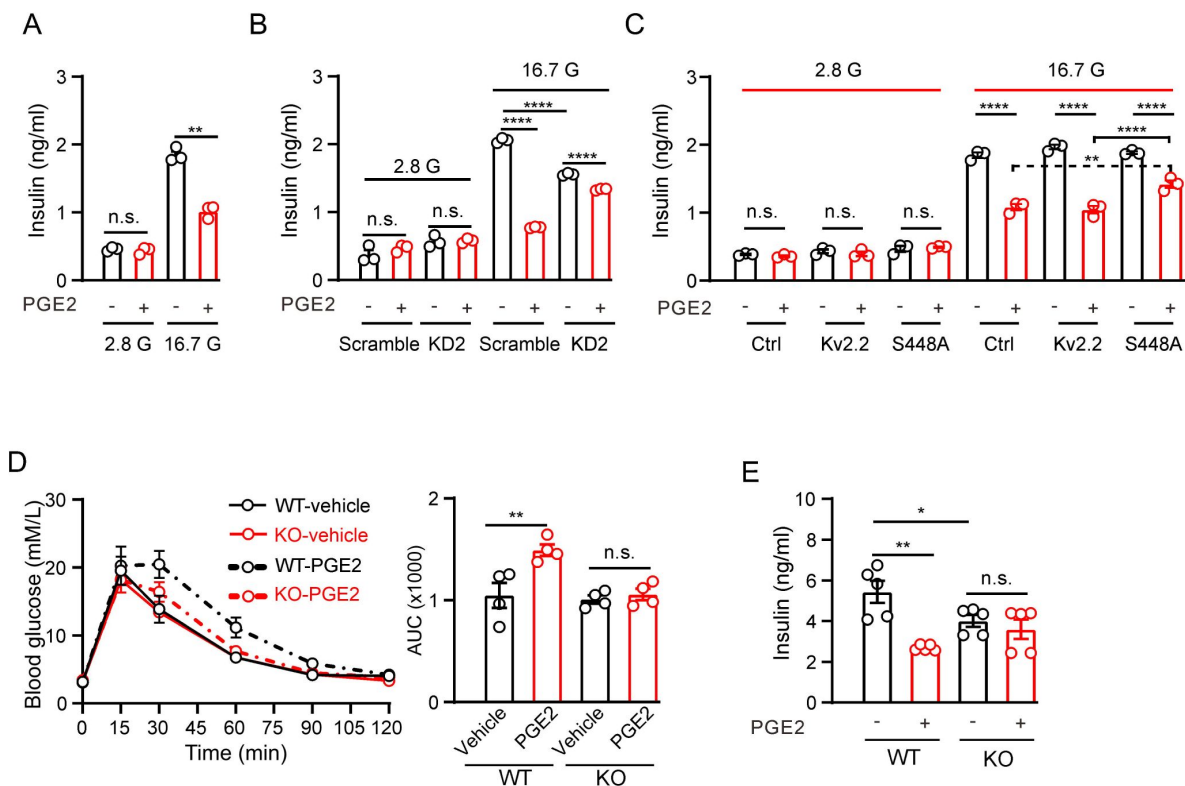


Figure 7.

PGE2 inhibits glucose-stimulated insulin secretion (GSIS) through Kv2.2 channels.

(A) Effects of PGE2 on insulin secretion in INS-1(832/13) cells under basal (2.8 mM, 2.8 G) or stimulatory (16.7 mM, 16.7 G) glucose concentrations ($n = 3$). 2.8 G, PGE2: n.s., not significant ($p = 0.9497$); 16.7 G, PGE2: $**p = 0.0082$. Two-tailed unpaired t -test. **(B)** Knockdown of Kv2.2 channels reduced GSIS and greatly alleviated the PGE2-induced inhibition of GSIS in INS-1(832/13) cells ($n = 3$). $****p < 0.0001$; n.s., not significant. Two-tailed unpaired t -tests. **(C)** Effects of PGE2 on insulin secretion in INS-1(832/13) cells transfected with empty vectors, Kv2.2, or Kv2.2-S448A mutant channels under basal (2.8 mM, 2.8 G) or stimulatory (16.7 mM, 16.7 G) glucose concentrations ($n = 3$). Overexpression of Kv2.2-S448A mutant channels greatly alleviated the PGE2-induced inhibition of GSIS in INS-1(832/13) cells. n.s., not significant; $****p < 0.0001$; $**p = 0.0096$. Two-tailed unpaired t -test. **(D)** Left, the effect of PGE2 on the glucose tolerance test in Kv2.2 $^{-/-}$, and control animals ($n = 4$ animals per group). Right, statistics for AUC from Left. $**p = 0.0064$. n.s., not significant ($p = 0.9684$). Two-tailed unpaired t -tests. **(E)** Statistics for the effect of PGE2 on GSIS (16.7 mM glucose) in isolated islets from Kv2.2 $^{-/-}$, and control animals. $N = 5$ animals per group. $*p = 0.0483$, $**p = 0.001$. n.s., not significant ($p = 0.4766$). Two-tailed unpaired t -tests.

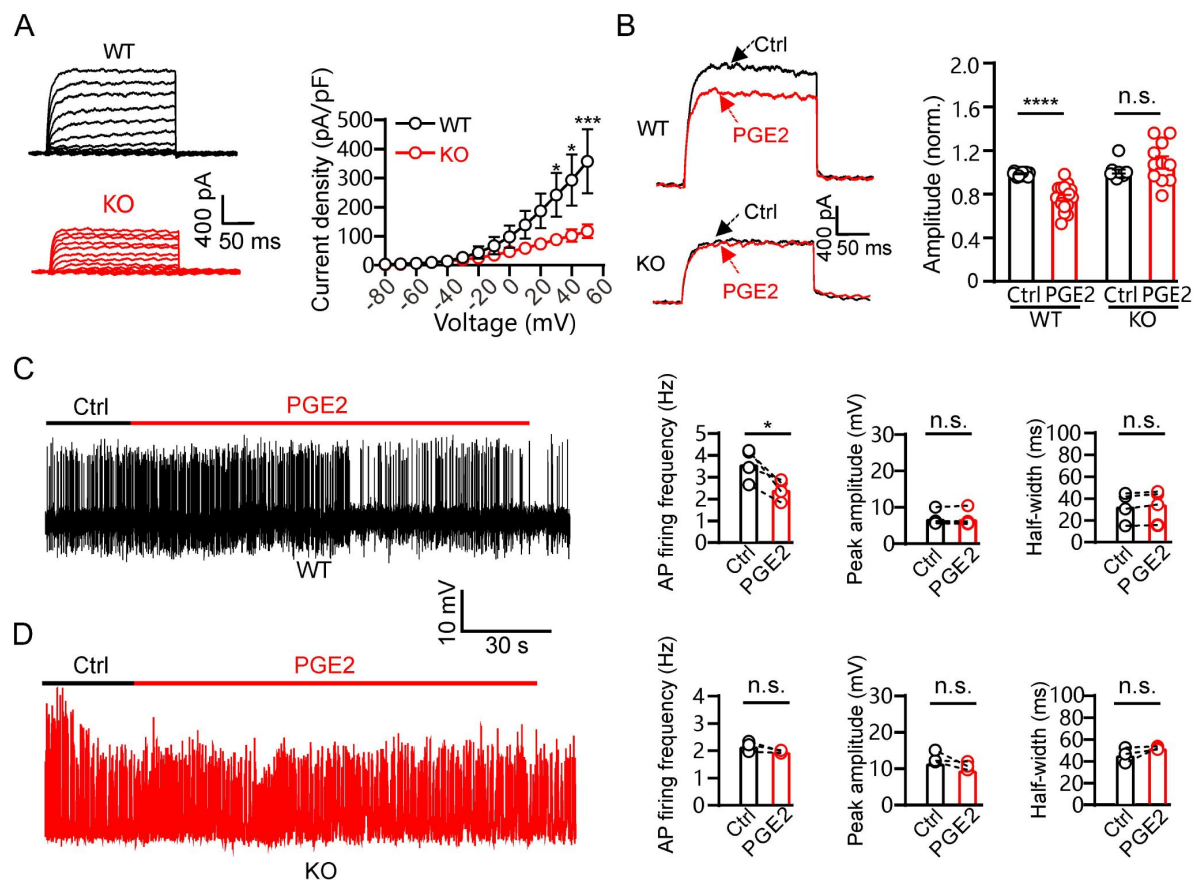


Figure 8.

PGE2 inhibits mouse pancreatic β -cell electrical excitability through Kv2.2 channels.

(A) Left, representative I_K recordings in response to 200-ms depolarization pulses from -80 to +50 mV in wild-type and Kv2.2^{-/-} knockout mouse pancreatic β -cells. Right, plot of the current-voltage relationship from Left ($n = 7$ for each data point). **(B)** Left, representative I_K traces induced by a depolarization pulse from -80 to +40 mV under the control condition and subsequently in the presence of 10 μ M PGE2 in the same β -cell from wild-type or Kv2.2^{-/-} mice. Right, statistics for the I_K amplitude using a two-tailed paired t -test ($n = 17$). **** $p < 0.0001$; n.s., not significant ($n = 12$, $p = 0.6317$). **(C)** Left, representative AP firings induced by 20 mM glucose under the control condition and subsequently in the presence of 10 μ M PGE2 in the same β -cell within wild-type mouse pancreatic islets. Right, statistics for the AP firing frequency (* $p = 0.0288$), amplitude ($p = 0.8589$), and half-width ($p = 0.0563$) from Left ($n = 4$). Two-tailed paired t -tests. **(D)** Left, representative AP firings induced by 20 mM glucose under the control condition and subsequently in the presence of 10 μ M PGE2 in the same β -cell within Kv2.2^{-/-} mouse pancreatic islets. Right, statistics for the AP firing frequency ($p = 0.1536$), amplitude ($p = 0.1981$), and half-width ($p = 0.2385$) from Left ($n = 3$). Two-tailed paired t -tests.

investigation, we unveil that Kv2.2 channels play a crucial role in regulating β -cell repetitive firing and insulin secretion. PGE2 inhibits Kv2.2 channels via the EP2/4 signaling pathway, consequently impeding GSIS in normal β -cells.

Even though the previous study reported that the EP2 protein-encoding gene *Ptger2* was not identified by RNA-seq in mouse islets (Ku et al., 2012 [\[1\]](#)). Other studies have shown that the mRNA expression of all four EPs is observed in rat, mouse, and human pancreatic islets (Bramswig et al., 2013 [\[2\]](#); Tran et al., 2002 [\[3\]](#); Vennemann et al., 2012 [\[4\]](#)). In our current study, we provide evidence demonstrating the presence of protein expression for all four types of PGE2 receptors in both human and mouse pancreatic β -cells. In comparison to EP3, our knowledge of the impact of EP1, EP2, and EP4 on insulin secretion is currently limited. Tran et al. have reported that the EP1 antagonist fails to impede the inhibitory effects of IL-1 β on GSIS in isolated rat islets. EP3 agonists, such as misoprostol or sulprostone, decrease GSIS through Gi proteins in rat islets (Tran et al., 2002 [\[3\]](#)). A recent study has unveiled that the activation of Gi/o protein-coupled receptors leads to the stimulation of Na⁺/K⁺ ATPase. This stimulation, in turn, hyperpolarizes the membrane potential of β -cells, consequently suppressing β -cell electrical excitability and insulin secretion (Dickerson et al., 2022 [\[5\]](#)). Consequently, one would anticipate that PGE2 activates Gi protein-coupled EP3 receptors, thereby inhibiting insulin secretion through membrane potential hyperpolarization and the reduction of β -cell electrical excitability. Surprisingly, no alterations in β -cell membrane potential were observed following PGE2 treatments. We found that PGE2 inhibits Kv2.2 channels and electroactivity of β -cells via G_s-coupled EP2/4 receptors instead of EP3 receptors. PGE2 is likely to predominantly exert its effects via the EP2/4 receptors in normal β -cells. However, in instances of established β -cell dysfunction, such as in T2DM, it appears that PGE2 may act through the upregulated EP3 receptors (Kimple et al., 2013 [\[6\]](#)). This study found that activating either EP2 or EP4 receptors can inhibit Kv2.2 channels, and further research is needed to determine which receptor is the primary regulatory factor for GSIS.

Both Kv2.1 and Kv2.2 contribute to the delayed outward K⁺ current in human β -cells, and their mRNA expression in diabetic islets is lower than that in non-diabetic islets (Fu et al., 2017 [\[7\]](#)). Kv2.1 channels in cell membrane exist in two forms: clustered and non-clustered. Non-clustered Kv2.1 channels conduct K⁺ normally, while clustered Kv2.1 channels are barely conductive (Fox, Loftus, & Tamkun, 2013 [\[8\]](#)). A recent study has shown that Kv2.1 channels form clusters in INS-1(832/13) cells and human β -cells (Fu et al., 2017 [\[7\]](#)). Kv2.1-mediated insulin exocytosis requires Kv2.1 clustering and a direct interaction with syntaxin 1A but is not dependent on its electrical function (Dai et al., 2012 [\[9\]](#); Fu et al., 2017 [\[7\]](#)). Consistent with the findings of the previous study (Jensen et al., 2013 [\[10\]](#)), we observed a significant reduction in GSIS in INS-1(832/13) cells upon knockdown of Kv2.2. Moreover, knockout of Kv2.2 also reduced GSIS in mouse islets. Kv2.2 channels play a crucial role in maintaining repetitive AP firing by promoting the recovery of voltage-gated sodium channels from inactivation (Johnston et al., 2008 [\[11\]](#)). Inhibition of Kv2.2 channels significantly reduces the repetitive firing in medial nucleus of the trapezoid body neurons and cortical pyramidal neurons (Johnston et al., 2008 [\[11\]](#); Wang et al., 2024 [\[12\]](#)). In the present study, we found that the major effect of PGE2-induced inhibition of Kv2.2 is to reduce the firing rate in β -cells, suggesting that Kv2.2 plays a key role in the β -cell repetitive firing upon stimulation. The PGE2-induced reduction in β -cell repetitive firing would decrease calcium channel opening, and thus, insulin secretion. Importantly, our findings pertain to physiological conditions, and while we demonstrate the inhibitory effects of PGE2 on Kv2.2 channels in normal β -cells, the role of this pathway under diabetic conditions requires further investigation and will be the focus of future studies.

In summary, this study uncovers a previously unknown role of the EP2/4-PKA-Kv2.2 signaling pathway in the inhibitory effect of PGE2 on GSIS in normal β -cells. This provides valuable insights into the complex interplay between prostaglandins, potassium channels, and insulin secretion, contributing to our understanding of pancreatic islet function and potential implications for diabetes mellitus.

Materials and methods

Ethics statement

All studies were carried out in strict accordance with the recommendations in the Guide for the Care and Use of Laboratory Animals of the National Institutes of Health. Animal experimental protocols were all approved by the Committee on the Ethics of Animal Experiments of Fudan University. The human pancreatic samples utilized in this study were sourced from the Biobank of Endocrine and Metabolic Diseases. All research procedures were approved by the Ethics Committee of Huashan Hospital, Fudan University, in accordance with the principles of the Helsinki Declaration.

Cell culture

Human embryonic kidney (HEK293T) cells were purchased from the cell bank of the Chinese Academy of Science. HEK293T cells were cultured in DMEM supplemented with 10% fetal bovine serum and 1% penicillin-streptomycin solution. INS-1(832/13) β -cells (Soltani et al., 2011) were cultured in RPMI 1640 with 10% fetal bovine serum, 1% penicillin-streptomycin solution and 0.1% β -mercaptoethanol. All the reagents were purchased from Thermo Fisher Scientific (Waltham, MA, USA).

Molecular biology

Plasmids for rat Kv2.2 (NM_054000.2) channels in pEGFPN1 vectors were as previously reported (Li et al., 2022). T13D, S14D, T17D, S367D, S448D, S710D, S799D, T801D, S840D, and S448A mutations of the Kv2.2 channel were achieved by PCR-based site-directed mutagenesis using ClonExpress Multis One Step Cloning Kit (Vazyme, Jiangsu, China). All mutations were confirmed by sequencing. HEK293T Cells were transiently transfected with wild-type or mutant Kv2.2 channels for 24 hours using Lipofectamine™ 2000 (Thermo Fisher Scientific, Waltham, MA, USA) before patch clamp recordings. For knockdown plasmids targeting Kv2.2, the shRNA hairpin sequences were inserted into BamHI and HindIII sites of the PAAV-shRNA targeting vector. Oligonucleotides specifying the shRNA are 5'-GGAGCAGATGAACGAAGAACT-3' (KD1-Kv2.2), 5'-GCTGGAGATGCTATACAATGA-3' (KD2-Kv2.2) and 5'-GCACCCAGTCCGCCCTGAGCAAA-3' (Scramble). Total RNA extraction and quantitative real-time polymerase chain reaction (qRT-PCR) were performed as previously described (T. Yang et al., 2016). Briefly, qRT-PCR was performed in 20 μ L reactions containing: 2 μ L of template, 0.4 μ mol/L of each paired primer, and SYBR Green PCR master mix. The thermo-cycling conditions were 94°C, 10 minutes; 40 cycles of 95°C, 30 s; 55°C, 30 s; 72°C, 60 s; and 72°C, 8 minutes. Results were normalized by β -actin mRNA. Primers were as previously reported (Hoshino et al., 2007): EP1(forward, reverse): 5'-ACCTTCTTTGGCGGCTCT and 5'-GCACGACACCACCATGATAC; EP2: 5'-CCACCTCATTCTCCTGGCTA and 5'-CGACAACAGAGGACTGAACG; EP3: 5'-AGCTTATGGGATCATGTGC and 5'-TCTGCTTCTCCGTGTGTGTC; EP4: 5'-TGCGAGTATTCGTCAACCAG and 5'-GGTCTAGGATGGGGTTCACA; β -actin: 5'-GGACTTCGAGCAAGAGATGG and 5'-AGCACTGTGTTGGCGTACAG.

Western blot

HEK293T cells were treated with PGE2 for 2, 5 and 10 minutes respectively. HEK293T cell homogenates were prepared using a lysis buffer (20 mM HEPES, 0.5% NP-40, 150 mM NaCl, 10% glycerol, 2 mM EDTA, 50 mM NaF, 0.1 mM Na₃VO₄) with protease inhibitor (P8340, Sigma, St. Louis, USA) and phosphatase inhibitor (P5726, Sigma, St. Louis, USA) cocktail. The protein samples were separated by 10% SDS-PAGE and then transferred to polyvinylidene fluoride membranes (1620177, BIO-RAD, Hercules, USA). The membranes were blocked with 10% non-fat dry milk in Tris-buffered saline with Tween-20 for 1 hour at room temperature and then incubated with primary

antibodies overnight at 4°C (anti-phospho-PKA, 1:1000, 5661, Cell Signaling Technology, Danvers, USA; anti-GAPDH, 1:1000; AG019, Beyotime, Shanghai, China). The blots were developed using enhanced chemiluminescence reagents and imaged using the ChemiDoc XRS⁺ imaging system from Bio-Rad (Hercules, CA, USA) and manufacturer's software.

Immunofluorescence

HEK293T and INS-1(832/13) cells were fixed in 4% paraformaldehyde (PFA) for 15 minutes, washed, and blocked (10% donkey serum, 1% BSA) for 2 hours at room temperature. Cells were then incubated in primary antibody solution (primary antibody (anti-EP1: 1:200, Ab217925, Abcam, Cambridge, UK; anti-EP2: 1:200, Ab167171, Abcam, Cambridge, UK; anti-EP3: 1:200, SC-57105, Santa Cruz Biotechnology, Santa Cruz, USA; anti-EP4: 1:200, Ab217966, Abcam, Cambridge, UK; anti-Kv2.2: 1:200, APC-120, Alomone Labs, Jerusalem, IL), 1% horse serum, and 0.3% Phosphate Buffered Saline with Tween 20 (PBST)) for 1 day at 4°C. Cells were washed three times with PBST and incubated overnight at 4°C in a secondary antibody solution containing Cy3-labeled goat anti-rat IgG (1:500, A0516, Beyotime, Shanghai, China) or Cy3/FITC-labeled goat anti-mouse IgG (1:500, A0521, Beyotime, Shanghai, China), 1% horse serum, and 0.3% PBST. Following another wash with PBST, the cells were treated with DAPI, and imaged using the Nikon A1⁺ Confocal Microscope System. For mouse and human pancreatic tissue, after overnight treatment in 4% PFA, the tissues were washed three times with PBS and then dehydrated in a fresh 30% sucrose solution. Subsequently, the pancreatic tissue was embedded in optimal cutting temperature compound and sliced into 25 µm sections using a cryostat. Slices were blocked with 10% horse serum and 0.3% PBST for 2 hours at room temperature. Following this, the slices underwent the same treatment and imaging procedure as INS-1(832/13) cells described above. The insulin antibody was purchased from Proteintech (1:1000, 66198-1-Ig, Wuhan, China). The specificity of the EP receptor antibodies was validated in INS-1(832/13) cells using knockdowns (Supplemental Fig. S2).

Electrophysiology

Whole-cell potassium currents in HEK293T cells and β-cells were recorded using a Multiclamp 700B amplifier (Molecular Devices, CA, USA). The extracellular solution contained (in mM): 140 NaCl, 2.5 KCl, 10 glucose, 2.5 CaCl₂, 10 HEPES, and 1 MgCl₂, pH 7.4 adjusted with NaOH. The pipette (2–3 MΩ) solution contained (in mM): 135 K-gluconate, 10 KCl, 1 CaCl₂, 1 MgCl₂, 10 HEPES, 2 Mg-ATP, and 10 EGTA, pH 7.3 adjusted with KOH. Whole-cell potassium currents were sampled at 10 kHz and filtered at 2 kHz. For β-cell action potential recordings, the extracellular solution contained (in mM): 119 NaCl, 4.7 KCl, 2 CaCl₂, 1.2 MgSO₄, 1.2 KH₂SO₄, 10 HEPES, pH 7.4 adjusted with NaOH. The pipette solution contained (in mM): 90 KCl, 50 NaCl, 1 MgCl₂, 10 EGTA, 10 HEPES, pH 7.3 adjusted with KOH. All of the electrophysiological recordings mentioned above were performed at room temperature.

Isolation of Primary Mouse Pancreatic Islets

Pancreatic islets were isolated from 7-week-old wild-type and KCNB2-KO male mice as previously described (Lernmark, 1974 [\[2\]](#)). Briefly, mice were euthanized, and a collagenase XI solution (0.5 mg/ml, C7657, Sigma, St. Louis, USA) was injected into the pancreas via the common bile duct with approximately 3 ml per mouse. The intact pancreas was carefully dissected, and digested for 16 min at 37°C. Afterward, 20 mL of HBSS solution was added, and the mixture was filtered through a 60-micron sieve. After brief centrifugation, a density gradient centrifugation system was established using Histopaque-1077 (10771, Sigma, St. Louis, USA), Histopaque-1119 (11191, Sigma, St. Louis, USA), and HBSS solution for pancreatic islet separation and purification. The islets were manually picked under a stereo microscope and cultured in RPMI 1640 with 10% fetal bovine serum, 1% penicillin-streptomycin solution. The time between islet isolation and the experiment typically ranged from 24–48 hours.

Intraperitoneal glucose tolerance test

Male mice were fasted overnight (16 hours) and then injected with D-glucose at a dose of 2.5 g/kg of body weight (intraperitoneal). Blood samples were collected from the tail vein before glucose injection ($t = 0$) and 15, 30, 60, 90, and 120 minutes after the glucose administration ($N = 4$ animals per group). A 20% glucose solution tailored to each mouse's actual body weight (following the standard of 10 $\mu\text{l/g}$) was intraperitoneally injected. Glucose levels were measured using a glucometer (Elite, Bayer, GER), following the manufacturer's recommendations. PGE2 (500 $\mu\text{g/kg}$ of body weight) or saline was intraperitoneally injected 1 hour before the glucose administration. The sampling of blood glucose level detection process was carried out in a double-blind fashion.

Insulin secretion assays

For GSIS in INS-1(832/13) cells: INS-1(832/13) cells were cultured in 24-well plates, grown to approximately 90% confluency, washed with PBS, and preincubated for 1 hour in KRB solution (in mM: 120 NaCl, 4.7 KCl, 2.5 CaCl_2 , 1.2 KH_2PO_4 , 1.2 MgSO_4 , 25 NaHCO_3 , 10 HEPES) without glucose. The cells were incubated in the KRB solution for an additional 1 hour in the presence of either 2.8 mM (2.8 G) or 16.7 mM (16.7 G) glucose with or without PGE2, as indicated. Insulin levels in KRB solution were quantified by the Insulin ELISA Kit (90080, Crystal Chem, Chicago, USA). For GSIS in islets: islets from 5 WT or KO male mice aged 7 weeks were used. Ten islets per mouse were preincubated in KRB buffer (supplemented with 0.025% BSA, pH 7.4) for 1 hour. Subsequently, the medium was removed, followed by sequential treatment with KRB solution (containing 16.7 mM glucose with or without PGE2) for 1 hour. Insulin in the supernatant was quantified using the same ELISA kit as above.

Generation of Kv2.2 knockout mice

The Kv2.2 channel coding gene, KCNB2 (Accession: NM_001098528.3), underwent conventional knockout using CRISPR-Cas9 gene editing. Two guide RNAs (gRNAs) were designed to target the KCNB2 gene. The sequences for the gRNAs are as follows: gRNA1: GAGAGTTAAGATCAACGTAG; gRNA2: AACTCGTCCGTGGCTGCAAA. The Cas9 protein was prepared and complexed with the two synthesized gRNAs to form ribonucleoprotein complexes. Ribonucleoprotein complexes were microinjected to fertilized C57BL/6 mouse oocytes to induce double-strand breaks at the KCNB2 gene target sites. The microinjected embryos were cultured in vitro to the appropriate developmental stage and then transferred into the oviducts or uteruses of pseudo pregnant recipient female mice. The recipient females gave birth to F_0 offspring. Genotyping of F_0 Mice: DNA was extracted from the tail tips of the F_0 mice. PCR was performed using two pairs of primers to amplify the regions flanking the target sites of the KCNB2 gene. The primer sequences are as follows: F1: TGATGTGGCGATGCCTATTCC; R1: TTCCACAGACTAACACTTACGG; R2: TCTTCTGATGGTATCTGGCTTGG. The PCR products were purified and sequenced to confirm the knockout of the KCNB2 gene. Mice confirmed to have the desired KCNB2 gene knockout were bred to produce a stable line of KCNB2 conventional knockout mice for further studies. The KCNB2 knockout mouse was created by Cyagen Biosciences (Suzhou, China). All animals were accommodated in specific pathogen-free Fudan University facilities following a 12-hour light-dark cycle.

Data analysis

The electrophysiological data were analyzed using Clampfit 10.7 (Molecular Devices, CA, USA). Quantitative analysis of the western blot experiments was performed with ImageJ (v1.53, NIH, USA). Data are given as the mean $\pm$ SEM. Two-tailed paired or unpaired t -test was used to compare two samples, and one-way ANOVA with Bonferroni *post hoc* test was employed for the comparison of multiple samples. A p -value < 0.05 was considered significant. All statistical analyses were performed using GraphPad Prism (v9.4, GraphPad Software Inc, USA).

Data Availability Statement

All data supporting the findings of this study are available within the paper and its Supplementary Information.

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Additional information

Author contributions

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Additional files

Supplemental Figure S1 and supporting text [🔗](#)

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Reviewer #2 (Public review):

The authors identified new target elements for prostaglandin E2 (PGE2) through which insulin release can be regulated in pancreatic beta cells under physiological conditions. In

vitro extracellular exposure to PGE2 could directly and dose-dependently inhibit the potassium channel Kv2.2. In vitro pharmacology revealed that this inhibition occurs through the EP2/4 receptors, which activate protein kinase A (PKA). By screening specific sites of the Kv2.2 channel, the target phosphorylation site (S448) for PKA regulation was found. The physiological relevance of the described signaling cascade was investigated and confirmed in vivo, using a Kv2.2 knockdown mouse model.

The strength of this manuscript is the novelty of the (EP2/4-PKA-Kv2.2 channel) molecular pathway described and the comprehensive methodological toolkit the authors have relied upon.

The introduction is detailed and contains all the information necessary to place the claims in context. Although the dataset is comprehensive and a logical lead is consistently built, there is one important point to consider: to clarify that the described signaling pathway is characteristic of normal physiological conditions and thus differs from pathological changes. It would be useful to carry out basic experiments in a diabetes model (regardless of in mouse or rat even).

Comments on revisions:

The authors addressed my comments sufficiently. I have no additional questions to clarify.

<https://doi.org/10.7554/eLife.97234.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public Review):

Summary:

This study investigated the mechanism by which PGE2 inhibits the release of insulin from pancreatic beta cells in response to glucose. The researchers used a combination of cell line experiments and studies in mice with genetic ablation of the Kv2.2 channel. Their findings suggest a novel pathway where PGE2 acts through EP2/EP4 receptors to activate PKA, which directly phosphorylates a specific site (S448) on the Kv2.2 channel, inhibiting its activity and reducing GSIS.

Strengths:

- *The study elegantly demonstrates a potential pathway connecting PGE2, EP2/EP4 receptors, PKA, and Kv2.2 channel activity, using embryonic cell line.*
- *Additional experiments in INS1 and primary mouse beta cells with altered Kv2.2 function partially support the inhibitory role of PGE2 on GSIS through Kv2.2 inhibition.*

Weaknesses:

- *A critical limitation is the use of HEK293T cells, which are not pancreatic beta cells. Functional aspects can differ significantly between these cell types.*
- *The study needs to address the apparent contradiction of PKA activating insulin secretion in beta cells, while also inhibiting GSIS through the proposed mechanism.*

- A more thorough explanation is needed for the discrepancies observed between the effects of PGE2 versus Kv2.2 knockdown/mutation on the electrical activity of beta cells and GSIS.

Thank you for your positive evaluation and constructive feedback on our study. We appreciate the concern regarding the use of HEK293T cells, which are not pancreatic beta cells and may exhibit functional differences. In response, we have repeated our key experiments using INS1 cells and primary mouse beta cells, which are more representative of the native beta cell environment. These additional experiments confirm our hypothesis and further support the role of Kv2.2 in PGE2-induced inhibition of GSIS. In beta cells, glucose-induced PKA activation is highly localized. As a result, while some PKA pathways promote insulin secretion, others may inhibit it. To directly demonstrate that PGE2-induced PKA phosphorylation of Kv2.2 is involved in the inhibitory effect on GSIS, we overexpressed the S448A mutant Kv2.2 channel in INS-1(832/13) cells. Our results show that Kv2.2-S448A channels significantly attenuate the inhibitory effect of PGE2 on GSIS, further supporting the critical role of Kv2.2 phosphorylation at S448. These data have been added to the revised Figure 7C.

Reviewer #2 (Public Review):

The authors identified new target elements for prostaglandin E2 (PGE2) through which insulin release can be regulated in pancreatic beta cells under physiological conditions. In vitro extracellular exposure to PGE2 could directly and dose-dependently inhibit the potassium channel Kv2.2. In vitro pharmacology revealed that this inhibition occurs through the EP2/4 receptors, which activate protein kinase A (PKA). By screening specific sites of the Kv2.2 channel, the target phosphorylation site (S448) for PKA regulation was found. The physiological relevance of the described signaling cascade was investigated and confirmed in vivo, using a Kv2.2 knockdown mouse model.

The strength of this manuscript is the novelty of the (EP2/4-PKA-Kv2.2 channel) molecular pathway described and the comprehensive methodological toolkit the authors have relied upon.

The introduction is detailed and contains all the information necessary to place the claims in context. Although the dataset is comprehensive and a logical lead is consistently built, there is one important point to consider: to clarify that the described signaling pathway is characteristic of normal physiological conditions and thus differs from pathological changes. It would be useful to carry out basic experiments in a diabetes model (regardless of whether this is in mice or rats).

Thank you for your positive evaluation and insightful comment. We have clarified in the Discussion section that our findings pertain specifically to physiological conditions. We acknowledge the importance of investigating the signaling pathway in a pathological context and plan to conduct experiments using a diabetes model in future studies to explore how this pathway may differ under such conditions.

Recommendations for the authors:

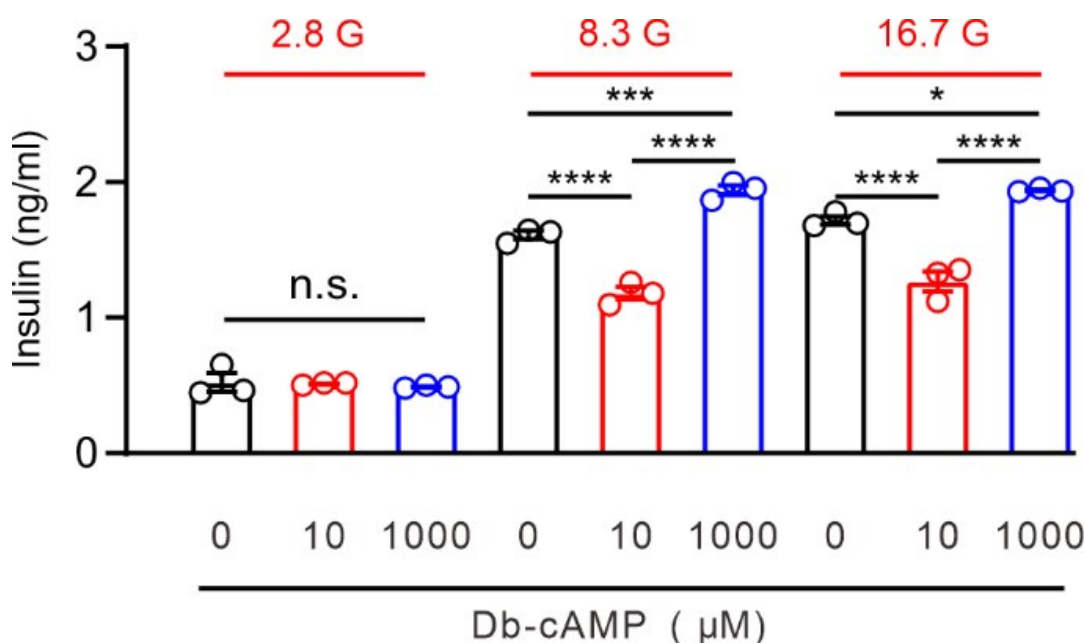
Reviewer #1 (Recommendations For The Authors):

(1) Figure 3A-C: PKA activation regulates different functional aspects in beta cells and HEK293T cells. It is well known that PKA activation enhances insulin secretion in beta cells, therefore the mechanisms that allow the same pathway at the same time to inhibit GSIS are not clear and should be addressed by experiments in beta cells.

Thank you for your insightful comment. Specificity and versatility in cAMP-PKA signaling are governed by the spatial localization and temporal dynamics of the signal. In beta cells, glucose-induced PKA activation is highly localized (Tengholm and Gylfe, 2017). As a result, while some PKA pathways promote insulin secretion, others may inhibit it. For example, a global increase in cAMP, such as through treatment with Db-cAMP, can simultaneously activate both stimulatory and inhibitory PKA pathways, reflecting a more integrated, complex response. In previous studies, 1 mM Db-cAMP was shown to enhance GSIS in INS-1 cells (Dezaki et al., 2011). We observed that 1 mM Db-cAMP increased GSIS, but lower concentrations (10 mM) decreased GSIS (as shown in Author response image 1). These findings suggest that not all PKA signaling events increase GSIS. To further investigate the role of PGE2-induced PKA phosphorylation of Kv2.2 in the inhibition of GSIS, we overexpressed the S448A mutant of Kv2.2 in INS-1 (832/13) cells. Our results showed that the Kv2.2-S448A mutant significantly attenuated the inhibitory effect of PGE2 on GSIS. These new data have been incorporated into the revised Figure 7C.

Author response image 1.

Effect of Db-cAMP on GSIS in INS-1 cells. Statistics for the effect of different concentrations of Db-cAMP on GSIS in INS-1(832/13) cells. One-way ANOVA with Bonferroni post hoc test. * $p < 0.05$; *** $p < 0.001$; **** $p < 0.0001$; n.s., not significant.



(2) Figure 3G: One would expect that the phospho-mimetic mutation, S448D, will have an opposite effect to S448A and a similar effect as PGE2 or PKA activator in Figure 3B. There is no explanation by the authors for having the same effect in S448A and S448D.

Thank you for your thoughtful comment. Indeed, the S448D mutation exhibited a similar effect to PGE2 on Kv2.2 channels, as we observed significantly smaller currents compared to wild-type Kv2.2 (Figure 3F). The S448D mutation mimics the phosphorylated state of S448, and since PGE2 regulates Kv2.2 channels by phosphorylating this residue, it has no further effect on the S448D mutant (Figure 3G). In contrast, the S448A mutation prevents phosphorylation at this site, which explains why PGE2 has no effect on the currents of S448A mutant Kv2.2 channels (Figure 3H). These results confirm that PGE2 modulates Kv2.2

channels specifically through phosphorylation of S448, as evidenced by the lack of effect on both the S448A and S448D mutants.

(3) Figure 4E: Since both PGE2 and Kv2.2 KD inhibit the activity of the channel, it doesn't definitively prove whether PGE2 acts through Kv2.2 in INS-1 cells. A complementary experiment should be done in which overactivation of Kv2.2 rescues the effect of PGE2. For example, with the S448A form of the channel.

We appreciate your comment and valuable suggestion. Knockdown of Kv2.2 abrogated the inhibitory effect of PGE2 on I_K currents in INS-1 cells (Figure 4E and F), which strongly indicates that PGE2 acts through Kv2.2. While we agree that the suggested complementary experiment with Kv2.2 overactivation (e.g., using the S448A mutant) could provide additional insights, we believe the current data sufficiently support our conclusion, as the knockdown of Kv2.2 eliminates the observed PGE2 effect, providing direct evidence of the channel's involvement.

(4) Figure 5C: This result requires further explanation. If PGE2 downregulates Kv2.2 activity and has an inhibitory effect on GSIS, why does Kv2.2 KD have the opposite effect?

The knockdown of Kv2.2 (Fig. 5C) reduced action potential (AP) firing rates compared to the scramble control (Fig. 5B), which is expected because Kv2.2 is critical for maintaining AP firing. When Kv2.2 is knocked down, the reduced AP firing diminishes the system's responsiveness to further modulation by PGE2. This is because PGE2 exerts its effects primarily through Kv2.2 channels. Therefore, in the Kv2.2 knockdown condition, PGE2 does not exert an additional inhibitory effect on AP firing rates, as the channels critical for its action are already impaired.

(5) Figure 5D - The EP1-EP4 receptor antibodies should be validated at least in INS-1(832/13) cells using knockdowns.

Thank you for your suggestion. We have validated the EP1-EP4 receptor antibodies in INS-1(832/13) cells using knockdown experiments. The validation results, including confirmation of specificity and knockdown efficiency, are provided in Supplemental Figure S2.

(6) Figure 7B - These experiments don't necessarily prove that PGE2 acts directly through Kv2.2 inhibition. Using the S448A mutation in these experiments could prove this point.

Thank you for this valuable suggestion. We have now overexpressed the S448A mutant Kv2.2 channels in INS-1(832/13) cells, and the results demonstrate that Kv2.2-S448A channels significantly reduce the inhibitory effect of PGE2 on GSIS. These new data have been incorporated into the revised Figure 7C.

Reviewer #2 (Recommendations For The Authors):

(1) Deficiencies and inaccuracies in the description of the methods (animal numbers, name of vendors, abbreviations) and the typos in the figures (axis label) require correction.

Thank you for pointing this out. We have carefully reviewed the manuscript and the figures, making the necessary corrections to address the deficiencies in the methods section and the typos in the figure axis labels.

(2) Reducing the number of figures (Figures 7/C-E: knockout mouse line test and Figure 1/HEK cell experiments could be part of supplementary) and paragraphs would

make the manuscript more compact and powerful. It would also ease its reading for non-experts.

Thank you for your suggestion. We have moved Figures 7C-E to the supplementary data (Supplemental Figure S1) to streamline the main manuscript.

(3) Multiple immunostainings for EP receptors in insulinoma cells or pancreatic islets would be representative.

Due to the rabbit-derived nature of the antibodies (EP1, EP2, EP4), performing multiple immunostainings on the same samples is not feasible due to potential cross-reactivity. However, the immunohistochemistry images demonstrate that each antibody labels more than 90% of the cells, indicating that β -cell express different subtypes of EP receptors simultaneously.

(4) The antagonists chosen (AH6809, AH23848) are non-specific. Experiments should be re-run (at least some) under more stringent conditions.

Thank you for your suggestion. AH6809 and AH23848 are well-documented, widely used antagonists in the literature. To further strengthen our findings, we have included additional, widely-used antagonists: the EP2-specific antagonist TG4155 and the EP4-specific antagonist GW627368. The results obtained with these new antagonists were consistent with those observed using AH6809 and AH23848. These updated data are now included in the revised Figure 4I and 4J.

(5) It would be very helpful to indeed emphasise that this work is for physiological conditions and that it is (or is not) modified in diabetes. Maybe even irrelevant for diabetes (?). This needs to be clarified and supported by data even if one could assume the authors intend to have a follow-up entirely dedicated to pathological changes, perhaps.

Thank you for this insightful comment. We have clarified in the Discussion that our findings are specific to physiological conditions. To address this point, we have added the following statement:

"Importantly, our findings pertain to physiological conditions. While we demonstrate the inhibitory effects of PGE2 on Kv2.2 channels in normal β -cells, the role of this pathway under diabetic conditions remains to be investigated and will be the focus of future studies."

Dezaki K, Damdindorj B, Sone H, Dyachok O, Tengholm A, Gylfe E, Kurashina T, Yoshida M, Kakei M, Yada T (2011) Ghrelin attenuates cAMP-PKA signaling to evoke insulinostatic cascade in islet β -cells. *Diabetes* 60:2315-2324.

Tengholm A, Gylfe E (2017) cAMP signalling in insulin and glucagon secretion. *Diabetes Obes Metab* 19 Suppl 1:42-53.

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